Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light Spectrum for Long Duration Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	george.brainard@jefferson.edu	**Fax:**	FY 215-923-7588
PI Organization Type:	UNIVERSITY	**Phone:**	215-955-7644
Organization Name:	Thomas Jefferson University		
PI Address 1:	Light Research Program		
PI Address 2:	1025 Walnut St., Room 507		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	09/01/2004	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	7
No. of PhD Candidates:	3	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smith, Peter (University of Arizona) Hanifin, John (Thomas Jefferson University) Pineda, Carissa (Thomas Jefferson University) Lockley, Steven (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HPF00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this project is to determine the optimal wavelengths of light for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits are considered a risk factor during space missions. The resulting physiological and behavioral changes from sleep and circadian disruption can threaten the success of a mission by diminishing alertness, cognitive ability, and psychomotor performance. Over 45% of all medications taken in space are sleep aids. Ambient light is the primary stimulus for circadian regulation in humans. While on Earth, humans have a 24-hour day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either a rapidly changing or severely disrupted day/night cycle, and must work in a dimly lit spacecraft interior with few or no windows. Bright white light has already been implemented as a pre-launch countermeasure, but has yet to be used during space flight. Providing sufficient light intensities to work areas in the International Space Station (ISS), the planned Crew Exploration Vehicle (CEV), and future lunar habitats raises several concerns, among which are heat production, energy consumption, and upmass. Mitigating these factors requires a better understanding of how different wavelengths of light regulate the human circadian system. From this data it may be possible to optimize the light spectrum both as a countermeasure for sleep and circadian disturbances before or during space exploration. Listed below are the specific aims of this project. Given changes in NASA priorities (Vision for Space Exploration) and subsequent budget cuts to this project, we re-aligned the third specific aim mid-project: 1) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Our published action spectrum indicates that the blue monochromatic wavelengths (446-477 nm) are the most potent for melatonin suppression. This aim will determine if increasing the power in the blue region of a broad spectrum, polychromatic fluorescent light can increase potency for melatonin suppression while keeping the visual experience white. The prototype blue-enriched lamps and exposure systems have been donated by Philips Lighting B.V., an NSBRI industry partner. This year over 95 nighttime melatonin suppression trials have been completed with human subjects (over 270 trials since the project's conception in 2004.) 2) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for circadian phase-shifting when they are enriched in the blue portion of the spectrum. For this aim, we will compare the magnitude of circadian phase-shifts in healthy subjects following exposure to equal photon densities of blue-enriched fluorescent lights versus the white lamps currently used by NASA. We initiated these trials in July 2007. 3) Test the hypothesis that three ambient extraterrestrial light environments (skylight during low earth orbit for ISS and the CEV, ambient light on the lunar surface, and ambient light on the surface of Mars) will differ in their capacity to regulate melatonin. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial and passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of light exposures in non-terrestrial environments. To test this hypothesis, we are working collaboratively with investigators from the Lunar and Planetary Laboratory (LPL) of the University of Arizona who are experts in characterizing the light in a variety of extraterrestrial environments. We will use a newly designed optical array to establish melatonin suppression responses with polychromatic light that simulates light in low Earth orbit, ambient light on the lunar surface and ambient light on the surface of Mars. The first prototype of this optical system used to simulate the spectral characteristics of light in extraterrestrial environments was delivered to our lab in March 2007. Human subject trials began immediately afterwards. In the coming year, we plan on completing subject light exposures, plasma melatonin assays, and data analysis for the fluence response curves for highly blue-enriched fluorescent lamps. If the data shows promise, these lamps may be used as a pre-launch countermeasure and also may provide NASA with a viable lighting countermeasure to sleep and circadian disruption for use during space exploration. We also plan to continue subject recruitment to determine the circadian phase-shifting ability of prototype "blue-enriched" fluorescent lamps. Finally, we will continue the melatonin suppression studies with the three extraterrestrial lighting simulations. Ultimately, these data will impact NASA lighting requirements and improve light treatment used as a countermeasure for circadian and sleep-wake disruption in space exploration. The data will also help NASA identify the best spectral transmission for space suit visors, space vehicles, and habitat windows, and the ideal spectral distribution for illumination of living quarters. If the trials using blue enriched polychromatic lamps succeed, these lamps may provide NASA with their first deliverable lighting countermeasure to sleep and circadian disruption for use during space exploration. Increasingly, however, it may be desirable to consider examining the potential utility of blue-enriched solid state luminaries for space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness to occur due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States alone, some 20 million Americans work shifts which do not allow them a biologically healthy nocturnal sleep cycle. This group has been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, cognitive and emotional problems. Furthermore, recent epidemiological studies of female night-shift nurses have shown that they are statistically more likely to suffer from breast cancer and colon cancer compared to day shift workers. Our laboratory is involved in testing the hypothesis that inappropriate exposure to light at night is a risk factor for cancer (Blask, et al., 2005). Aside from evidence for a breakdown in physical health, the effects of circadian disruption and sleep loss have long been known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of psychomotor coordination and physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military, as well. Additionally, many people experience the same effects due to air travel across several time zones. Understanding the biological impact of ambient light on the human circadian system is a critical step toward the eventual application of countermeasures to these and other sleep and circadian-related disorders in civilian populations. Quantifying the relative potency of different wavelengths of light for melatonin suppression via the retinohypothalamic tract will provide key insights not only into the basic underlying physiology of this unique photosensory system, but also

	produce guidelines for how a misalignment of the system could be most effectively corrected. This research is also expected to have a wide-ranging impact on future types of artificial indoor lighting and architectural illumination in general. The lighting industry has taken notice of the biological impact different types of light is capable of having, and has already begun to invest in furthering our understanding of light's role for human health. There are a number of existing therapeutic interventions using light that stand to benefit from an enhanced understanding of how specifically different wavelengths of the spectrum affect the human circadian system. A more efficient intervention, with increased potency and/or less side-effects could result. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD). Similar bright white light interventions are also used for treating jetlag. Side effects from exposure to bright white light for this and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders could deliver the same medical impact with lower levels of light intensity, and potentially fewer side-effects. Our group has initiated Phase I testing of light therapy with blue solid-state lighting for SAD patients (Glickman, et al., 2006).
Task Progress:	This year we have completed the plasma melatonin assays and data analysis for fluence response studies for the 4,000 Kelvin and 17, 000 Kelvin Philips lamps and have begun modeling work on the curve-fitting of this data. In addition, a majority of the subject exposures for the "special blue" lamp have been completed and the assays and data analysis will be completed over the coming year. The modeling for all three fluence response curves will be completed once the assays and data analysis for the "special blue" lamp has been completed. These three fluence response curves will fulfill Specific Aim 1: testing the relative circadian potency of the 17,000 Kelvin and "special blue" lamps in relation to the 4,000 Kelvin lamp, which is similar to the current white polychromatic fluorescent lamps currently used by NASA for pre-launch countermeasures. We completed renovation of the laboratory to allow us to conduct 7 day, in-patient phase-shifting studies with the 4,000 Kelvin and 17,000 Kelvin lamps. In collaboration with Stephen Lockley, Ph.D. of Brigham and Women's Hospital, a phase-shifting protocol was developed and lab personnel were trained. Subject recruitment and screening was initiated and is ongoing. Two male subjects were run through the protocol in July 2007, and more subjects are scheduled for two runs later this summer. Specimens from the first run have not been assayed yet. Data from these studies will contribute to fulfilling Specific Aim 2. For Specific Aim 3, we have collaborated with the LPL of the University of Arizona to develop a prototype optical system that can simulate ambient light in extraterrestrial environments. The design and fabrication of the prototype, however, has been more challenging than anticipated. A working model was completed during this past year, and installed in our laboratory in March 2007. We have initiated subject recruitment and screening for the melatonin suppression fluence response study with a simulation of lunar light. Over 20 nights of simulated lunar light exposure have already been completed. The remainder of the lunar light fluence response study is ongoing. Plasma melatonin assays collected in these studies have not yet been performed. We are now working with the LPL to acquire two more light systems to increase subject throughput for Aim 3.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Abstracts for Journals and Proceedings	Brainard GC, Hanifin JP, Rollag MD. "Photoreception for circadian, neuroendocrine and neurobehavioral regulation." 8th International Congress of Physiological Anthropology, Kamakura, Japan, October 9-14, 2006. 8th International Congress of Physiological Anthropology, October 2006. , Oct-2006
Abstracts for Journals and Proceedings	Brainard GC, Rollag M, Warfield B, Stone M, Martin E, Fucci R, Byrne B, Jasser S, Durkan B, James M, Gerner E, Hanifin JP. "Fluence-response curve for melatonin suppression with blue-enriched fluorescent light." 18th Annual Meeting of the Society for Light Treatment and Biological Rhythms, Quebec City, Canada, July 13-15, 2006. 18th Annual Meeting of the Society for Light Treatment and Biological Rhythms, July 2006. , Jul-2006
Abstracts for Journals and Proceedings	Brainard GC, Warfield B, Fucci R, Martin E, Stone M, James M, Byrne B, Rollag M, Hanifin JP. "Testing polychromatic light as a stimulus for space travel." 19th Annual Meeting of the Society for Light Treatment and Biological Rhythms, Copenhagen, Denmark, June 28-30, 2007. 19th Annual Meeting of the Society of Light Treatment and Biological Rhythms Program and Abstracts. 2007 June;19:2. , Jun-2007
Abstracts for Journals and Proceedings	Brainard GC, Warfield B, Martin E, Stone M, Fucci R, James M, Durkan B, Hanifin JP, Byrne B, Rollag MD. "Optimizing light as a countermeasure in space exploration." NASA Human Research Program Investigators Workshop, League City, Texas, February 12-14, 2007. NASA Human Research Program Investigators Workshop, February 2007. , Feb-2007
Abstracts for Journals and Proceedings	Brainard GC. "Light and health: photoreception for biological, behavioral and therapeutic responses in humans." 26th Session of the CIE, Beijing, China, July 4-11, 2007. 26th Session of the CIE. 2007 July;2:D6-2-D6-3. , Jul-2007
Abstracts for Journals and Proceedings	Brainard GC. "Photoreception for the neurobehavioral effects of light in humans." 2nd CIE Expert Symposium on Light and Health, Ottawa, Canada, September 7-8, 2006. 2nd CIE Expert Symposium on Light and Health, September 2006. , Sep-2006
Abstracts for Journals and Proceedings	Gooley JJ, Brainard GC, Rajaratnam SM, Kronauer RE, Czeisler CA, Lockley SW. "Spectral sensitivity of the human circadian timing system." 5th World Congress of the World Federation of Sleep Research and Sleep Medicine Societies, Cairns, Australia, September 2-6, 2007. Sleep and Biological Rhythms 2007 Aug;5 Suppl 1:A22. http://dx.doi.org/10.1111/j.1479-8425.2007.00283.x , Aug-2007
Articles in Peer-reviewed Journals	Hanifin JP, Brainard GC. "Photoreception for circadian, neuroendocrine, and neurobehavioral regulation." J Physiol Anthropol. 2007 Mar;26(2):87-94. PMID: 17435349 , Mar-2007

Articles in Peer-reviewed Journals	Jasser SA, Hanifin JP, Rollag MD, Brainard GC. "Dim light adaptation attenuates acute melatonin suppression in humans." J Biol Rhythms. 2006 Oct;21(5):394-404. PMID: 16998159 , Oct-2006
Awards	Brainard GC. "Invited Keynote Speaker at the 2nd International Commission on Illumination Expert Symposium on Light and Health, Ottawa, Canada, September 7, 2006." Sep-2006
Awards	Brainard GC. "Invited Keynote Speaker at the Light and Wellbeing Symposium, London, United Kingdom, September 27, 2006." Sep-2006
Awards	Brainard GC. "Invited to teach a master's class at Lightfair International, New York, New York, May 9, 2007." May-2007
Awards	Smith P. "Leader of the current Phoenix Mars Mission, scheduled to launch August 3, 2007, and will be the first Mars mission run from a university." Aug-2007
Papers from Meeting Proceedings	Brainard GC, Provencio I. "Photoreception for the neurobehavioral effects of light in humans." 2nd CIE Expert Symposium for Light and Health, Ottawa, Ontario, Canada, September 6-8, 2006. Paper from the 2nd CIE Expert Symposium proceedings, 2006, p. 6-21. , Sep-2006
Significant Media Coverage	Johnson ML. "Interview with G. Brainard discussing the significance of the circadian photoreceptor system findings on residential lighting. NSBRI and NASA research was described during the interview, but not in final article, although general discussion in interview with Brainard." Article, "Retinal Revolution," in ResidentialLighting.com, May 1, 2007. http://residentiallighting.com/IntheMagazine/ArticleLanding/tabid/83/Default.aspx?tid=1&cid=6785&issueid=347 , May-2007
Significant Media Coverage	Kreykenbohm S, Gronbach R. "Interview regarding the influence of light on human physiology and the future of research in the area. Article specifically describes NASA and NSBRI research." Light Live! 2007 Jan;8(1):4-9., Jan-2007